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Review Report

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Contents

1. Manuscript Summary

This paper by Jani et al. addresses a critical challenge in acute myeloid leukemia (AML) treatment, where patients exhibit highly variable responses to therapeutic agents despite similar clinical presentations. AML's biological heterogeneity, characterized by diverse genetic mutations and molecular subtypes, makes personalized treatment selection essential for improving patient outcomes. Current approaches rely primarily on retrospective association analyses that cannot predict individual patient responses to specific drugs.

The authors developed PREDICT-AML, a machine learning framework that integrates multi-omics data from the BeatAML cohort to predict personalized drug responses. Their approach combines transcriptomic profiles, somatic mutations, clinical annotations, and engineered drug representations including molecular fingerprints and computed drug-pathway distances. The framework employs multiple machine learning algorithms, with CatBoost emerging as the optimal model through comprehensive benchmarking across 165 unique drugs and 520 patient samples.

The optimal model achieved cross-validation performance of r=0.85±0.02 and MAE=21.02±1.5, though test set performance showed substantial degradation to r=0.66 and MAE=39.04. Feature importance analysis revealed that drug molecular fingerprints, drug-pathway distances, and specific gene expression patterns ($CD_{300}E$, $HNRNPA_2B_1$) were primary drivers of predictions. The study identified mechanistically relevant associations, particularly between Venetoclax response and immunogenic cell death pathway activity (R^2=0.637), and MEK inhibitors with conventional dendritic cell-like signatures.

This work represents an important step toward precision medicine in AML by demonstrating the feasibility of multi-modal machine learning for treatment selection, though the performance degradation highlights remaining challenges for clinical implementation.

2. Review Synopsis

This study presents PREDICT-AML, a machine learning framework for predicting personalized drug responses in acute myeloid leukemia using multi-omics data from the BeatAML cohort. While the work addresses an important clinical need and demonstrates substantial technical effort in feature engineering, several critical methodological concerns require attention to ensure the validity of the predictive framework and its clinical utility.

The most significant issue concerns the temporal train/test partitioning strategy, which uses collection Waves 1+2 for training and Waves 3+4 for testing without adequate validation of potential systematic biases between collection periods. The reliance on visual t-SNE inspection to dismiss batch effects is methodologically insufficient, and the absence of wave-specific demographic analyses, formal batch effect testing, or alternative partitioning validation strategies undermines confidence in the generalization claims. This temporal confounding could explain the substantial performance degradation observed from cross-validation to test set.

The cross-validation methodology requires strengthening to address potential data leakage arising from the hierarchical structure of patient-drug combinations. With multiple drug responses per patient among 34,387 patient-drug records, the 5-fold cross-validation strategy must respect patient-level dependencies to prevent optimistic bias. The dramatic performance drop from cross-validation (r=0.85, MAE=21.02) to independent testing (r=0.66, MAE=39.04) suggests fundamental validation flaws that could compromise model reliability.

Additionally, the evaluation framework lacks essential baseline comparisons against fundamental prediction strategies such as mean response, drug-specific averages, or patient-specific averages. Without these critical benchmarks and appropriate null model validation through permutation testing, the clinical advancement represented by the complex machine learning approach remains inadequately demonstrated.

3. Major Comments

3.1 Cross-Validation Methodology and Data Leakage

Hierarchical Data Structure in Cross-Validation Design

Critique:

The cross-validation methodology requires careful examination to address potential data leakage arising from the hierarchical structure of patient-drug combinations. With 34,387 patient-drug records from 520 samples representing multiple drug responses per patient, the 5-fold cross-validation strategy must respect patient-level dependencies to prevent optimistic bias (Varoquaux et al., 2016). If drug responses from the same patient appear in both training and validation folds, this creates information leakage that could explain the substantial performance degradation from cross-validation (r=0.85±0.02, MAE=21.02±1.5) to independent test set (r=0.66, MAE=39.04)—an 86% increase in error that strongly suggests overfitting (Arlot & Celisse, 2009). The cross-validation standard deviations based on only 5 folds provide unreliable uncertainty estimates for robust model selection decisions.

Relevant section:

> *Section 2.6 "Machine Learning Models" mentions 5-fold cross-validation but lacks specification of patient-level versus sample-level splitting.*

Framework for addressing:

- Implement patient-level cross-validation where all drug responses for a given patient are assigned to the same fold to prevent information leakage between training and validation sets
- Stratify folds to maintain balanced representation of key patient characteristics and drug classes across validation splits
- Increase cross-validation folds to 10-fold or implement repeated k-fold cross-validation to improve reliability of uncertainty estimates
- Report fold-specific performance metrics to assess stability and identify potential outlier folds that could indicate validation problems
- Conduct permutation tests on cross-validation results to validate that performance exceeds chance levels given the high-dimensional feature space

3.2 Model Generalization and Overfitting Assessment

Systematic Overfitting Detection and Regularization

Critique:

The substantial performance degradation observed across multiple models from cross-validation to test set requires systematic investigation to address potential overfitting issues that could compromise clinical utility. The feedforward neural network's dramatic performance drop (CV MAE=21.02 to test MAE=41.96, representing 95% degradation) exemplifies classic overfitting behavior, yet the authors provide insufficient analysis of underlying causes or mitigation strategies (Hawkins, 2004). The evaluation of only 15 novel drugs (9.8% of test drugs) provides inadequate statistical power for robust generalization assessment to truly unseen compounds, which is crucial for clinical deployment (Sheridan, 2013). The absence of nested cross-validation creates fundamental validation flaws when hyperparameter optimization uses the same folds as performance estimation.

Relevant section:

Table 1 shows consistent CV-to-test performance drops across models, and Section 3.1 mentions novel drugs without separate analysis.

Framework for addressing:

• Implement learning curve analysis to examine model convergence and identify optimal training set sizes that balance bias-variance trade-offs
• Conduct systematic regularization parameter optimization using nested cross-validation to prevent optimistic performance estimates
• Analyze performance separately for the 15 novel drugs versus 138 previously seen drugs to properly assess generalization to new chemical entities
• Apply early stopping criteria and complexity penalties appropriate for each model class to mitigate overfitting in high-dimensional settings
• Examine feature importance stability across different training subsets to identify potentially spurious predictive relationships that may not generalize
• Report confidence intervals for all performance metrics using bootstrap resampling to properly quantify prediction uncertainty

3.3 Model Selection and Hyperparameter Optimization

Systematic Hyperparameter Tuning Methodology

Critique:

The model comparison framework requires substantial strengthening through rigorous hyperparameter optimization procedures to ensure fair algorithm evaluation. The complete absence of any systematic tuning methodology—no mention of grid search, random search, or Bayesian optimization—fundamentally undermines the validity of model selection conclusions (Bergstra & Bengio, 2012). Tree-based algorithms like CatBoost, XGBoost, and LightGBM have numerous critical hyperparameters affecting regularization, model complexity, and generalization that significantly impact performance when properly tuned (Probst et al., 2018). The suspiciously consistent performance across deep learning architectures (GAT, CNN, LSTM all achieving MAE ~40.7-40.85) suggests inadequate optimization rather than genuine algorithmic similarities.

Relevant section:

Section 2.6 "Machine Learning Models" lacks any description of hyperparameter optimization procedures.

Framework for addressing:
• Implement nested cross-validation with an outer loop for performance estimation and inner loop for hyperparameter optimization to prevent optimistic bias in performance estimates
• Apply systematic hyperparameter tuning for each algorithm using either grid search over predefined ranges or Bayesian optimization for efficiency in high-dimensional parameter spaces
• Document specific hyperparameter ranges searched and final optimal values selected for each algorithm to ensure reproducibility
• Include regularization parameter optimization for all models, particularly learning rates, tree depth, and complexity penalties for gradient boosting methods
• Conduct statistical significance testing of performance differences between optimized models using appropriate resampling procedures or permutation tests to validate model selection decisions

3.4 Performance Validation and Baseline Comparisons

Fundamental Baseline Model Implementation

Critique:
The evaluation framework requires strengthening through implementation of essential baseline models that are critical for contextualizing the utility of complex machine learning approaches. The absence of fundamental baselines such as mean response prediction, drug-specific averages, or patient-specific averages makes it impossible to assess whether the reported CatBoost performance (MAE=39.04, r=0.66) represents meaningful improvement over trivial prediction strategies . Given the 0-300 drug response range, a simple mean baseline would provide crucial interpretative context for the MAE magnitude. The lack of null model validation through permutation testing is particularly concerning in high-dimensional settings where spurious correlations can emerge from chance relationships between 2,333 features and limited samples .

Relevant section:

> *Section 3.1 "Benchmarking Machine Learning Models" lacks any baseline model comparisons.*

Framework for addressing:
• Implement fundamental baseline models including global mean prediction, drug-specific mean responses, and patient-specific mean responses to establish minimum performance thresholds
• Conduct permutation testing by randomly shuffling response labels while maintaining feature structure to validate that observed correlations exceed chance levels
• Compare against established methods using identical train/test splits, specifically implementing ElasticNet regression and other published approaches on the same data partitions
• Test individual feature modality models (clinical-only, genomic-only, drug-only) to quantify incremental benefits of multi-modal integration
• Validate statistical significance of performance improvements over baselines using appropriate resampling procedures and confidence interval estimation

3.5 Statistical Reporting and Methodological Transparency

Multiple Testing Correction and Statistical Significance

Critique:

The statistical reporting framework requires substantial enhancement to meet rigorous standards for biomedical machine learning applications, particularly regarding multiple testing corrections and cross-validation methodology specification. The feature association analysis claims statistical significance ($q<0.05$) across thousands of features and multiple drugs without clearly specifying the multiple testing correction method employed, creating substantial risk of false discoveries (Benjamini & Hochberg, 1995). The substantial CV-to-test performance degradation (MAE increase from ~21 to 39, correlation drop from ~0.85 to 0.66) represents a critical limitation that requires prominent emphasis rather than being understated as demonstrating "generalization capability" (Kapoor & Narayanan, 2023). Critical methodological details regarding patient-level versus sample-level cross-validation splitting remain unspecified, preventing reproducibility assessment.

Relevant section:

Section 3.5 mentions q<0.05 without correction details; Section 2.6 lacks CV specification; Abstract understates performance degradation.

Framework for addressing:

• Explicitly specify multiple testing correction methods (FDR, Bonferroni) used for feature association analyses and report both raw and corrected p-values with clear justification for chosen approaches
• Prominently discuss the 85% error increase from CV to test performance in abstract, results, and discussion as a key limitation affecting clinical applicability rather than minimizing this concerning finding
• Clarify cross-validation methodology by specifying whether patient-level or sample-level splitting was employed and provide detailed procedural descriptions for reproducibility
• Report confidence intervals for all performance metrics using appropriate resampling methods to quantify prediction uncertainty and enable proper interpretation of results
• Include systematic evaluation of novel drug performance separately from previously seen drugs to demonstrate true generalization capability essential for clinical deployment

3.6 Temporal Study Design and Data Partitioning

Temporal Confounding Assessment in Wave-Based Data Splitting

Critique:

The temporal train/test partitioning strategy requires more rigorous validation to ensure the robustness of generalization claims. The use of Waves 1+2 for training and Waves 3+4 for testing introduces potential systematic biases if collection periods differed in patient selection criteria, treatment protocols, laboratory procedures, or clinical contexts (Varoquaux et al., 2016). The sole reliance on t-SNE visualization to claim "no visible segregation between training and test sets" is methodologically insufficient, as t-SNE is a stochastic, non-linear technique that can mask subtle but important systematic differences (Wattenberg et al., 2016). Furthermore, the absence of wave-specific demographic breakdowns, formal batch effect testing, and alternative partitioning validation strategies undermines confidence in the temporal splitting approach.

Relevant section:

Section 2.1 "Data Collection and Processing" and Section 2.2 "RNA-Seq & DNA-Seq Analysis" where t-SNE analysis is described.

Framework for addressing:

• Conduct formal batch effect detection using methods such as principal component analysis variance decomposition or ComBat algorithms to quantitatively assess systematic differences between waves
• Report detailed wave-specific demographic and clinical characteristics including age distributions, disease subtypes, prior treatments, and performance status with statistical comparisons
• Implement rigorous statistical tests for batch effects including permutation-based approaches and systematic analysis of technical variables across collection periods
• Validate temporal partitioning by comparing performance against random patient-level splits and stratified sampling approaches to demonstrate that temporal splitting does not artificially inflate or deflate performance estimates
• Provide separate analysis of the 15 novel drugs in the test set versus drugs present in training to assess true generalization to unseen chemical entities
• Report time intervals between collection waves and document any changes in clinical protocols, laboratory procedures, or patient selection criteria that occurred between collection periods

4. Minor Comments

4.1 Biological Mechanism Validation

Drug-Pathway Association Mechanistic Validation

Critique:

The biological interpretation of identified drug-feature associations requires more rigorous mechanistic validation to distinguish genuine therapeutic relationships from spurious correlations. The strong negative correlations between MEK inhibitors (Selumetinib R^2=0.661, Trametinib R^2=0.683) and conventional dendritic cell-like signatures lack adequate mechanistic explanation, as MEK inhibitors primarily target the MAPK/ERK pathway in cancer cells rather than directly affecting immune cell differentiation states (Samatar & Poulikakos, 2014). The identified biomarker genes ($CD_{300}E$, $LGALS_2$, VCAN, $ARHGEF_{10}L$) exhibit remarkably high correlations (R^2 0.6-0.7) with multiple drugs, which is unusual for genuine biomarkers and suggests potential technical confounding, batch effects, or indirect associations rather than direct mechanistic relationships (Geeleher et al., 2014). The drug-pathway distance approach using STRING networks assumes network proximity translates to functional effects but lacks validation against established pharmacological databases or functional assays.

Relevant section:

> *Section 3.5 reports feature associations and Figure 6 shows regression analyses but lacks mechanistic validation.*

Framework for addressing:

• Validate drug-pathway distances by comparing with established drug-target databases and demonstrating that drugs with known similar mechanisms cluster appropriately
• Provide literature-supported mechanistic explanations for major associations, particularly MEK inhibitor-immune signature relationships
• Investigate whether high-correlation biomarkers represent technical artifacts by examining their behavior across different batches or processing groups
• Cross-reference identified genes with established AML drug resistance mechanisms and provide biological pathway context for observed associations

4.2 Methodological Transparency and Reproducibility

Computational Implementation Details and Parameter Specification

Critique:

The computational methodology requires enhanced documentation to ensure reproducibility and enable independent validation of the proposed framework. Critical implementation details including random-walk restart parameters, convergence criteria, and normalization procedures are insufficiently specified throughout the manuscript (Peng, 2011). The title contains misleading terminology, referring to "Drug Interaction" when the study examines individual drug responses rather than drug-drug interactions, potentially confusing readers about the study scope and methodology. Additionally, the comparison with previous methods (MDREAM, QPOP, MSDRP) lacks quantitative rigor through head-to-head evaluation on identical datasets with consistent evaluation metrics (Hutson, 2018).

Relevant section:

Title, Section 2.4 lacks parameter details, and Section 4 Discussion provides incomplete method comparisons.

Framework for addressing:

• Revise title to accurately reflect focus on "drug response prediction" or "drug sensitivity prediction" rather than "drug interaction"
• Specify all computational parameters including restart probability (e.g., 0.15), convergence tolerance (e.g., 1e-6), and maximum iterations for network propagation
• Provide complete software versions, random seeds, and hyperparameter settings used for each algorithm to enable exact reproduction
• Implement quantitative comparison with published methods using identical train/test splits and evaluation protocols rather than literature comparisons

4.3 Model Interpretability and Biological Validation

Feature Importance Stability and Interaction Analysis

Critique:

The interpretability analysis requires strengthening through systematic validation of feature importance stability and comprehensive interaction detection to ensure reliable biomarker identification. While SHAP values provide valuable individual prediction explanations, the analysis relies solely on additive decomposition without employing SHAP interaction values or alternative interaction detection methods to capture synergistic or antagonistic relationships between molecular features that could be critical for understanding drug response mechanisms (Lundberg et al., 2020). The consistency between SHAP-derived rankings and native CatBoost variable importance scores (Supplementary Figure 3) is not systematically evaluated, and different top features across models (CatBoost, RF, LightGBM) suggest potential instability in biomarker identification that requires statistical validation (Strobl et al., 2007).

Relevant section:

Figure 5 and Supplementary Figure 3 show feature importance analysis but lack stability validation or interaction detection.

Framework for addressing:

• Implement bootstrap analysis across cross-validation folds to assess feature importance stability and identify consistently important features versus fold-specific artifacts
• Apply SHAP interaction values to detect feature synergies and antagonistic relationships that may be masked by additive explanations
• Reconcile differences between SHAP and native importance rankings by investigating which method provides more reliable biological interpretation
• Validate top features against established AML biomarker literature and provide mechanistic rationales for identified gene expression patterns

4.4 Network Biology and Feature Engineering

GSVA Normalization Strategy and Feature Integration

Critique:

The differential normalization scheme applied to pathway activities ([-0.5, 0.5]) versus cell state enrichments ([0,1]) requires biological and statistical justification to prevent systematic bias in multi-modal feature integration. This arbitrary scaling creates fundamentally different distributions and variance structures that could disproportionately weight certain feature types in machine learning models (Hänzelmann et al., 2013). The combination of oncogene expression with the 150 most variable genes creates potential double-counting bias, as oncogenes may overlap with highly variable genes, artificially inflating importance of overlapping features without deduplication strategies (Ritchie et al., 2015).

Relevant section:

> *Section 2.2 describes GSVA scoring with different normalization ranges but lacks justification.*

Framework for addressing:

• Implement standardized z-score normalization across all enrichment scores to ensure equal variance contribution to machine learning models
• Conduct overlap analysis between oncogenes and the 150 most variable genes, reporting intersection statistics and implementing feature deduplication
• Test sensitivity of model performance to different normalization approaches and report results
• Provide biological rationale for differential scaling or adopt consistent normalization methodology

STRING Database Threshold Selection and Justification

Critique:

The drug-pathway distance calculation methodology requires more rigorous justification for the STRING protein-protein interaction network threshold selection. The choice of score ≥700 representing "strong experimental evidence" lacks adequate support through comparative analysis or literature precedent (Szklarczyk et al., 2018). In network biology applications, thresholds of 400-500 are commonly employed for high-confidence interactions, while 700+ represents very stringent criteria that may exclude biologically relevant indirect regulatory relationships critical for drug mechanism studies (Luck et al., 2020). The random-walk with restart parameters including restart probability, convergence criteria, and iteration limits are completely absent from the methodology, preventing reproducibility and optimization validation.

Relevant section:

> *Section 2.4 "Drug Pathway Distance Estimation" mentions the ≥700 threshold but provides no justification or parameter specifications.*

Framework for addressing:

• Conduct comparative analysis testing multiple STRING confidence thresholds (400, 500, 600, 700, 800) and report drug-pathway distance correlation with known drug-target relationships
• Specify all random-walk parameters including restart probability (typically 0.1-0.3), convergence tolerance (e.g., 1e-6), and maximum iterations
• Validate the approach by demonstrating that drugs with known similar mechanisms cluster appropriately in the computed distance space
• Provide literature citations supporting the chosen threshold and compare performance against established drug-target databases like DrugBank

4.5 Performance Evaluation and Clinical Interpretation

Clinical Decision Threshold Establishment

Critique:

The clinical interpretation framework requires enhancement through establishment of meaningful decision thresholds that contextualize prediction accuracy within AML treatment decision-making requirements. The reported MAE of 39.04 on a 0-300 AUC scale (approximately 13% relative error) and correlation of r=0.66 lack contextualization against clinical decision-making standards or existing prediction tools used in oncology practice . The aggregation of performance across 150+ drugs may mask critical drug-specific performance heterogeneity where some compounds may be highly predictable while others exhibit essentially random behavior, obscuring clinically relevant utility patterns (Collins et al., 2015).

Relevant section:

> *Section 3.1 reports aggregate performance metrics without clinical contextualization or drug-specific heterogeneity analysis.*

Framework for addressing:

• Establish clinically meaningful sensitivity/resistance thresholds based on AML treatment guidelines and report classification accuracy at these cutpoints
• Conduct drug-specific performance analysis reporting the range and distribution of prediction accuracy across the drug panel
• Compare performance against existing clinical prediction tools or standard-of-care approaches where available
• Implement decision curve analysis to quantify clinical utility at different decision thresholds relevant to treatment selection scenarios

Cross-Validation Variance Estimation and Stability Assessment

Critique:

The cross-validation uncertainty estimates require more rigorous statistical validation to ensure reliable model performance assessment. The reported standard deviations (e.g., r=0.85±0.02, MAE=21.02±1.5) appear implausibly small for biomedical prediction tasks involving high-dimensional data with inherent biological variability (Bischl et al., 2012). With 5-fold cross-validation yielding approximately 67 samples per fold from 337 training samples, such tight confidence intervals suggest potential methodological issues including inadequate variance estimation or hidden stratification factors not accounted for in the splitting strategy (Krstajic et al., 2014). The absence of fold-specific performance analysis prevents detection of unstable learning patterns or data leakage that could compromise model reliability.

Relevant section:

> *Table 1 reports cross-validation standard deviations but lacks fold-specific analysis or variance validation.*

Framework for addressing:

• Report individual fold performance metrics to assess variance patterns and identify potential outlier folds indicating validation problems
• Implement bootstrap resampling within cross-validation to generate more robust confidence intervals for performance estimates
• Conduct repeated k-fold cross-validation (e.g., 10 repetitions of 5-fold CV) to improve reliability of uncertainty quantification
• Compare variance estimates against established benchmarks from similar biomedical prediction tasks to validate plausibility

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